

HELSINKI KAISANIEMI, Finland

WMO: 029780

Lat: 60.1753N

Lon: 24.9447E

Elev: 13

StdP: 14.69

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -0.8	(c) 4.7	(d) -5.8	(e) 4.0	(f) -0.4	(g) 0.4	(h) 5.6	(i) 6.0	(j) 20.5	(k) 29.9	(l) 18.7	(m) 29.7	(n) 7.3	(o) 40	(p) 0.598

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.0	78.6	67.9	75.4	66.0	72.7	63.7	69.8	76.2	67.8	73.8	65.3	71.3	7.9	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
67.4	100.7	73.9	65.3	93.5	72.1	62.7	85.1	69.1	33.8	76.2	32.1	73.6	30.2	71.6	74.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 18.6	(b) 16.6	(c) 14.9	(e) -3.9	(f) 82.1	(g) 7.9	(h) 4.5	(i) -9.6	(j) 85.3	(k) -14.3	(l) 87.9	(m) -18.7	(n) 90.4	(o) -24.5	(p) 93.7		
(4) 18.6	(5) 16.6	(6) 14.9	(7) -4.4	(8) 70.4	(9) 7.8	(10) 3.0	(11) -10.0	(12) 72.5	(13) -14.5	(14) 74.2	(15) -18.9	(16) 75.9	(17) -24.6	(18) 78.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	44.2	23.8	26.3	31.4	40.7	52.0	58.9	65.1	62.6	54.9	43.9	37.8	31.6
	DBStd	15.43	10.06	9.81	6.54	5.43	6.86	5.29	5.28	4.63	5.32	6.04	7.00	8.58
	HDD50	3553	813	663	577	285	55	4	0	0	20	200	365	571
	HDD65	7708	1278	1083	1042	730	406	194	65	101	304	654	815	1036
	CDD50	1426	0	0	0	5	116	270	467	390	167	11	0	0
	CDD65	106	0	0	0	0	2	11	67	26	0	0	0	0
	CDH74	424	0	0	0	0	7	34	306	77	0	0	0	0
	CDH80	46	0	0	0	0	0	2	39	5	0	0	0	0
(14) Wind	WSAvg	8.1	8.4	8.3	8.6	7.5	7.5	7.8	7.4	7.4	7.9	8.5	9.0	9.5
(15) Precipitation	PrecAvg	23.7	1.7	1.3	1.2	1.3	1.4	2.0	2.2	2.9	2.4	2.7	2.6	2.0
	PrecMax	33.0	3.3	3.9	2.2	2.8	2.9	4.9	6.0	6.5	6.1	7.0	6.7	5.0
	PrecMin	14.2	0.0	0.0	0.2	0.3	0.2	0.4	0.1	0.0	0.7	0.6	0.2	0.2
	PrecStd	4.0	0.9	0.9	0.6	0.7	0.8	1.1	1.6	1.6	1.3	1.4	1.2	1.1
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.2	43.1	48.2	62.0	74.5	78.0	83.6	80.5	68.7	57.6	51.1	47.3
		MCWB	40.0	37.6	42.1	48.9	59.4	64.6	68.3	69.6	61.4	52.6	49.0	45.4
	2%	DB	39.6	39.3	44.3	57.3	70.7	73.8	80.4	75.8	65.8	55.3	48.8	44.5
		MCWB	38.3	36.8	38.3	47.6	57.5	62.0	68.1	67.7	60.2	52.2	47.2	42.5
	5%	DB	38.0	37.6	41.4	53.1	67.4	71.0	77.4	72.8	64.1	53.9	47.2	42.6
		MCWB	36.9	35.9	37.3	44.9	56.2	60.5	67.7	64.5	59.4	51.4	45.7	40.8
	10%	DB	36.0	36.2	38.9	49.7	63.8	68.3	74.6	70.2	62.4	52.3	46.0	41.0
		MCWB	35.0	35.0	35.8	42.5	54.2	58.8	65.9	62.9	58.2	49.9	44.5	39.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.5	38.7	42.2	50.4	63.2	67.6	72.0	71.3	63.0	54.8	49.6	46.0
		MCDB	40.8	40.8	46.9	59.1	70.0	73.5	78.9	76.9	66.0	56.5	50.7	46.8
	2%	WB	38.5	37.5	39.4	47.8	59.8	64.3	70.6	68.7	61.7	53.6	47.7	42.9
		MCDB	39.4	39.1	42.6	55.5	66.6	72.1	77.2	74.6	64.7	54.8	48.7	44.0
	5%	WB	37.1	36.1	37.9	45.9	57.6	62.0	69.2	66.6	60.5	52.0	46.1	41.1
		MCDB	38.1	37.3	40.6	52.3	65.3	68.9	75.5	71.4	63.2	53.4	46.9	42.3
	10%	WB	35.2	35.1	36.4	43.7	55.0	59.9	67.1	64.1	59.1	50.3	44.8	39.7
		MCDB	36.0	36.1	38.6	48.5	63.5	66.3	73.4	68.9	61.6	52.0	45.8	41.0
(35) Mean Daily Temperature Range	5% DB	MDBR	8.1	7.4	9.2	11.3	12.9	12.2	12.0	11.1	9.7	8.5	6.1	6.7
		MCDBR	5.7	6.5	11.1	16.7	17.3	15.9	14.6	13.9	11.1	9.1	6.0	6.5
		MCWBR	5.6	5.4	7.2	9.3	8.9	8.1	6.9	6.9	7.0	7.1	6.0	6.4
	5% WB	MCDBR	5.7	6.1	9.7	15.2	15.6	14.5	13.0	11.5	10.0	7.5	6.0	6.1
		MCWBR	5.7	5.4	6.9	9.1	8.8	8.2	6.8	6.6	7.0	7.0	6.1	6.1
(40) Clear-Sky Solar Irradiance	taub		0.280	0.313	0.335	0.360	0.358	0.356	0.378	0.372	0.345	0.334	0.307	0.265
	taud		2.141	2.192	2.247	2.340	2.414	2.477	2.438	2.454	2.519	2.394	2.216	2.091
	Ebn at Noon		150	212	249	266	276	278	268	260	249	207	138	106
	Edh at Noon		14	23	30	33	33	31	32	29	23	19	13	10
(44) All-Sky Solar Radiation	RadAvg		78	259	732	1220	1667	1732	1671	1295	789	344	96	47
(45)	RadStd		14	50	94	128	180	135	152	133	102	54	17	7

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (60 neighbors)	+0.76	N/A	N/A	N/A	N/A	N/A	N/A	-252	N/A	N/A	N/A

Nomenclature: See separate page